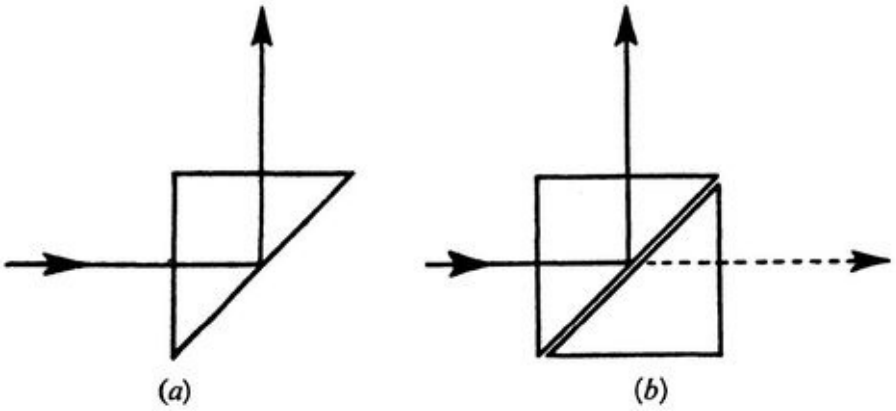


Figure 22.—Total reflection in a glass prism.

If a second prism is placed near the first one (Figure 22 (*b*)), no change will take place so long as the distance between them is sufficiently great. But as soon as this distance becomes very small, a little light can pass into the second prism, too. For in total reflection, a small quantity of light energy always seeps through the reflecting surface, but only for a very short distance, of the order of magnitude of the wavelength. If the second prism is brought sufficiently close, the light which has seeped through can penetrate into the second prism and can travel further in the ordinary manner. The closer the two surfaces are to each other, the greater is the quantity of light that goes through, and if the surfaces are pressed firmly together, no more total reflection occurs at all.

Something quite analogous to this phenomenon takes place in the case of the matter waves of the alpha particles. In this analogy, the interior is to be regarded as the equivalent of one prism, and the outside space as the equivalent of the other one, while the potential barrier between them is the equivalent of the layer of air separating the two prisms. Some waves will always get through the barrier, and the thinner is this barrier, the larger proportion of the waves will escape into the outer space. In this connection, ‘barrier’ means that part of the potential curve which rises above the horizontal energy level of the particle. From this it follows automatically that the higher is this level—and hence, the greater is the energy of the alpha particle—the more transparent will be the barrier to the waves of the alpha particle, for the potential barrier that must be overcome, will be proportionately narrower. Thus, assuming that originally the waves were present in the interior only, as more and more time elapses, we will find an increasingly larger portion of them outside, too, and will find that this portion increases with the increase of the energy of the particle.

Now this description in terms of the wave aspect must be translated back into the language of the particle aspect. In doing so, we must bear in mind that in the course of our discussion of the conditions prevailing within the